Challenge: Let's Chat

Narrative

Deep within the noise of routine transmissions, secrets travel in plain sight. A monitoring station somewhere on the network is broadcasting a steady stream of text - nothing unusual on the surface. But buried in the silence between each character, encoded in the rhythms of time itself, is a hidden message meant for no one in particular. Or perhaps for someone who knows exactly where to look.

Your team has intercepted the coordinates of this station, concealed behind layers of encoding and encryption. Decode them, connect, and listen carefully. The cosmos rewards those who pay attention to what others overlook.

Objective

The goal of this challenge is to connect to a chat server and receive an **overt** message. Note that chat server specifics will be provided at the beginning of the challenge. In the delays between each character of this message (beginning with the delay between the first and second characters) is a **covert** message. Of course, you already knew this. *Note that when you first connect, the server will send a prompt. You must respond with your team name followed by a newline before the transmission begins. Do not record timing data until after you have sent your team name.*

Once connected to the chat server (using a simple socket connection), you should expect continuous incoming data that is terminated with the string "EOF" – at which point the connection will be closed. The incoming data is, of course, the **overt** message. You must "extract" the **covert** message from the timings.

Having trouble getting the covert message? Set a debug mode that prints out the delays in between characters of the overt message and try to find a pattern (i.e., are there two unique delays?); subsequently, map the delays to 0 and 1 (or vice versa). Make sure not to consider any delays that represent the time between connecting to the server and receiving the first character of the message. Any decoded byte that proves problematic can be represented as a question mark in the message. It is possible that delays in the network require performing this process a number of times to decode the complete covert message.

When you think that you have successfully completed the challenge, send your **fully** decoded and decrypted covert message to your **team text channel** and @ **prof**.

Once you are done, **submit compressed folder with following files on Canvas**:

(1) A document that provides:
   (a) the **fully** decoded covert message; and
   (b) thorough documentation of your team process as you proceeded through the challenge, including description of each team member contributions;
(2) Any modified Python code (e.g., your chat program, Vigenere program, Binary program); and
(3) Any other scripts or programs that you created to assist you (if applicable).

You will be graded on how well you:

(1) Decode and decrypt the chat server IP and port (15%);
(2) Obtain, decode, and decrypt the covert message (60%);
(3) Document your process (20%); and
(4) Submit the required ZIP file (5%).

GOOD LUCK!